

$$\phi(r) = (r^2 + r_0^2)^{1/2} \quad (3.7.5)$$

where r_0 is a scale factor that you get to choose. Multiquadrics are said to be less sensitive to the choice of r_0 than some other functional forms.

In general, both for multiquadrics and for other functions, below, r_0 should be larger than the typical separation of points but smaller than the “outer scale” or feature size of the function that you are interpolating. There can be several orders of magnitude difference between the interpolation accuracy with a good choice for r_0 , versus a poor choice, so it is definitely worth some experimentation. One way to experiment is to construct an RBF interpolator omitting one data point at a time and measuring the interpolation error at the omitted point.

The inverse multiquadric

$$\phi(r) = (r^2 + r_0^2)^{-1/2} \quad (3.7.6)$$

gives results that are comparable to the multiquadric, sometimes better.

It might seem odd that a function and its inverse (actually, reciprocal) work about equally well. The explanation is that what really matters is smoothness, and certain properties of the function’s Fourier transform that are not very different between the multiquadric and its reciprocal. The fact that one increases monotonically and the other decreases turns out to be almost irrelevant. However, if you want the extrapolated function to go to zero far from all the data (where an accurate value is impossible anyway), then the inverse multiquadric is a good choice.

The thin-plate spline radial basis function is

$$\phi(r) = r^2 \log(r/r_0) \quad (3.7.7)$$

with the limiting value $\phi(0) = 0$ assumed. This function has some physical justification in the energy minimization problem associated with warping a thin elastic plate. There is no indication that it is generally better than either of the above forms, however.

The Gaussian radial basis function is just what you’d expect,

$$\phi(r) = \exp(-\frac{1}{2}r^2/r_0^2) \quad (3.7.8)$$

The interpolation accuracy using Gaussian basis functions can be very sensitive to r_0 , and they are often avoided for this reason. However, for smooth functions and with an optimal r_0 , very high accuracy can be achieved. The Gaussian also will extrapolate any function to zero far from the data, and it gets to zero quickly.

Other functions are also in use, for example those of Wendland [6]. There is a large literature in which the above choices for basis functions are tested against specific functional forms or experimental data sets [1,2,7]. Few, if any, general recommendations emerge. We suggest that you try the alternatives in the order listed above, starting with multiquadrics, and that you not omit experimenting with different choices of the scale parameters r_0 .

The functions discussed are implemented in code as: